

IEEE SPS Industrial Innovation Award

Paste-Ready Nomination Fields for Rahul Gupta

Prepared for Prof. Chi-Chun Lee (Jeremy) for the IEEE Signal Processing Society online nomination system.

1. Nominee Details

Award: IEEE Signal Processing Society Industrial Innovation Award

Nominee: Rahul Gupta

Current role: Senior Applied Science Manager, Responsible AI, Amazon AGI / Frontier AI Assets

Professional affiliation: Amazon

Location: Cambridge, Massachusetts, USA

Email: rahul.1987iit@gmail.com

Phone: (213) 587-1103

LinkedIn: <https://www.linkedin.com/in/rahul-gupta-16139818>

Google Scholar: <https://scholar.google.com/citations?user=gSTLsFcAAAAJ>

Education: PhD, Electrical Engineering, University of Southern California; B.Tech., Electrical Engineering, IIT Kharagpur

Suggested nomination title: For industrial innovation and academic ecosystem building in Responsible AI

2. Endorsement Fields

Endorser A: Volkan Cevher, EPFL. Email to confirm: volkan.cevher@epfl.ch

Endorser B: Kush R. Varshney, IBM Fellow, IBM Research. Email to confirm: krvarshn@us.ibm.com

3. Proposed Award Citation

For industrial innovation and academic ecosystem building in Responsible AI.

4. Short Nomination Summary

Rahul Gupta is a senior industry leader whose work has translated Responsible AI research into deployed industrial systems at Amazon while also helping build a broader academic research ecosystem around trustworthy AI. He has led Responsible AI programs spanning evaluation and safety enhancement for Alexa, Amazon Nova, and other initiatives in the Amazon AGI division. His teams have built foundational capabilities for large-scale conversational AI and frontier foundation models, connected technical risk evaluation to executive product decisions, and produced externally visible outcomes including benchmarks, workshops, academic challenges, patents, and peer-reviewed publications. Rahul has supported academic research through workshops and challenges such as TrustNLP (<https://tinyurl.com/26ftbw8j>), Trustworthy Speech Processing (<https://tinyurl.com/29wvcfmg>), the SemEval Unlearning Challenge (<https://tinyurl.com/23bf3t68>), and the Nova Trusted AI Challenge (<https://tinyurl.com/2danx9xk>), and through Amazon Academic Research Award collaborations (<https://tinyurl.com/2ym554fn>) collectively representing more than \$20M in funded research to academics.

5. Specific Accomplishments Associated with the Nomination and Citation

Industrial leadership accomplishments: Rahul Gupta's accomplishments center on building and leading Responsible AI programs inside Amazon. He is the overall head of Responsible AI and safety for Amazon's AGI / Frontier AI Assets organization, reporting directly to Vice President-level leadership. In Amazon Nova and AGI, he has led teams as large as 50+ across Nova products, such as Nova LMs, image generators, video generators, and the Nova Trusted AI Challenge. He also led Responsible AI work supporting Nova model release readiness under Amazon's Frontier Model Safety Framework (<https://tinyurl.com/24ksxzw5>) and, for the same, sought and provided regular technical feedback to external standards and evaluation bodies such as NIST, the Frontier Model Forum, and MLCommons. Amazon Science's *Building trust into AI* article (<https://tinyurl.com/28s24hr1>) identifies Rahul as the RAI lead for Amazon's AGI organization and describes the end-to-end pipeline his organization leads across pretraining, post-training, evaluation, and frontier-risk assessment. Previous to his role at Amazon AGI, Rahul founded and led the Responsible AI science team for Amazon Alexa, focused on private, fair, and trustworthy customer experiences in

large-scale conversational AI. He also served as a technical lead for large-scale NLU model training and testing.

Academic ecosystem-building accomplishments: Rahul’s accomplishments also include formal organizer and program-lead roles that support Responsible AI research outside Amazon. He helps organize the TrustNLP workshop series, and in the past organized Trustworthy Speech Processing @ ICASSP’24, and the SemEval Unlearning Challenge in 2025. These venues create recurring forums, shared tasks, and benchmarks for academic and industry researchers working on trustworthy language and speech processing. Rahul also led the Nova Trusted AI Challenge, for which hundreds of universities applied and the program has supported 10 universities for safety research year over year. He has served as an Amazon scientific point of contact for Academic Research Award collaborations collectively representing more than \$20M in funded academic research. His technical contributions include BOLD for bias measurement (<https://tinyurl.com/27vjlc89>), FLIRT for feedback-loop red-teaming (<https://tinyurl.com/2ya3cpny>), fairness methods for large-scale NLU systems (<https://tinyurl.com/28qyyr2a>), and CBRN uplift evaluation for frontier language models (<https://tinyurl.com/24lna944>). He has 25+ patented or patent-pending technologies and publications across NeurIPS, ICLR, ACL, EMNLP, FAccT, ICASSP, and other venues. He has also disseminated this work through public Nova safety reports, Amazon Science technical blogs on catastrophic-risk certification for LLMs (<https://tinyurl.com/26sd8vj2>) and trustworthy machine learning at Alexa AI (<https://tinyurl.com/2aqnvjff>), external Nova safety coverage, and invited talks or panels at IEEE SLT 2026 (<https://tinyurl.com/2435b596>), AdvML-Frontiers x CoTMA at COLM 2026 (<https://tinyurl.com/2der5ng9>), and Foundation Models for Social Good at IJCAI-ECAI 2026 (<https://tinyurl.com/22e9h814>).

6. Impact of the Contributions and Evidence Supporting the Claims

Industrial impact and evidence: Nova models have been adopted by tens of thousands of customers, in part attributable to the strong safety profile built by Rahul’s team. Nova has also been endorsed by multiple third-party evaluators as a top-line safety model. Amazon Science’s *Independent evaluations demonstrate Nova Premier’s safety* (<https://tinyurl.com/2y7yqgp3>) reports black-box stress testing and red-team exercises by evaluators such as PRISM Eval and ActiveFence. The Nova Premier critical-risk evaluation (<https://tinyurl.com/25hagozh>) further states that Nova Premier was safe for public release under Amazon’s Frontier Model Safety Framework. External coverage from ABNewswire (<https://tinyurl.com/2252v6bz>) reported that Anthropic and Amazon led in independent quantitative AI safety testing, while Quantum Zeitgeist (<https://tinyurl.com/26qau3vy>) summarized Nova Lite safety evaluation work as a robust FMSF assessment with third-party validation.

His patent portfolio provides additional evidence of Responsible AI concepts becoming production mechanisms. For example, during his Alexa leadership period, the technology reflected in *Cohort determination in natural language processing* (<https://tinyurl.com/275fb62t>) helped identify underperforming cohorts in NLP systems and use that signal to improve model performance across user groups. The technology was deployed for millions of Alexa customers, contributing to improved experiences for minority and underrepresented customer groups; it is one example from Rahul’s technical contributions.

Academic impact and evidence: The academic impact is visible through public venues and programs that create reusable research artifacts and bring universities into frontier-model safety work. Rahul’s Google Scholar profile (<https://tinyurl.com/27tungke>) reports close to 5,000 citations, reflecting broad research uptake across speech, NLP, and Responsible AI. The TrustNLP workshop series, Trustworthy Speech Processing, and the SemEval Unlearning Challenge provide venues and shared tasks for research on trustworthy language and speech processing. As an example of workshop leadership he has helped build, a 2026 analysis of six years of TrustNLP (<https://tinyurl.com/22ua8qn8>) documents the workshop’s growth from 8 archived proceedings papers in 2021 to 41 in 2026, with 144 proceedings papers across six editions. The Nova Trusted AI Challenge connected frontier model safety questions to university teams after hundreds of universities applied, and the program has supported 10 universities for safety research year over year. Together, these programs show that Rahul’s work has shaped not only deployed systems but also the research ecosystem that studies and evaluates Responsible AI.

7. Brief Biography or Link to CV

Rahul Gupta is a Senior Applied Science Manager focused on Responsible AI in Amazon’s AGI / Frontier AI Assets organization. His work spans safety, evaluation, red-teaming, fairness, privacy, and mitigation capabilities for frontier foundation models, Amazon Nova, and other initiatives in the Amazon AGI division. He received his PhD in Electrical Engineering from the University of Southern California and his B.Tech. in Electrical Engineering from IIT Kharagpur.

At Amazon Alexa, Rahul founded and led a Responsible AI science team that built foundational capabilities for private, fair, and trustworthy customer experiences for millions of Alexa customers. In Amazon AGI and Nova Responsible

AI, he has led teams as large as 50+ across safety, external evaluation, LMArena performance, robust guardrails for enterprise-scale Nova model adoption, and university AI safety challenges. He has 25+ patented or patent-pending technologies and publications across NeurIPS, ICLR, ACL, EMNLP, FAccT, ICASSP, and other venues, and helps organize TrustNLP, Trustworthy Speech Processing, and the SemEval Unlearning Challenge.